

Review Comments for MS 25325 “Information provision and externalities: The impact of genomic testing on the dairy industry”

The authors investigated the extent to which the adoption of genomic testing has affected the inbreeding rates of the U.S. dairy industry. They hypothesized that cattle breeders, to compete for market shares, are more likely to line breed bulls there were already popular before this technology became available in 2008 and less likely to line breed those other lines. Based on this argument, they used a matching procedure to identify the treatment and control groups of dairy bulls based upon comparable traits but with different popularity. Then they applied a difference-in-difference approach to estimate the average treatment effects on the treated (ATT). They reported an increase of one to 2.8 percentage points in inbreeding rates of the popular lines since 2012. Furthermore, they compared the benefits, in terms of increased milk production, to the costs, in terms of losses due to inbreeding, of adopting genomic testing by cattle breeders and reported a net benefit of 6.9-10.42 billion dollars for the U.S. dairy industry.

This is a novel application of well-established econometric approaches (i.e., matching and DID) to the U.S. dairy genetics market, a research area that I do not specialize in. I have the following comments from the methodological perspective and wonder whether the authors can elaborate more on the related backgrounds for the broad audience and, if necessary, improve their methodology, in a revised manuscript.

A key underlying assumption of this study is, based upon information from genomic testing, cattle breeders intentionally line breed already popular bulls to grab market shares for maximal profits. This begs several questions:

1. In the U.S. dairy industry, what is the approximate proportion of commercial dairy breeders who specialize in breeding and sell bull semen as the main business? And what is the proportion of dairy farmers who breed (most of) their cattle and raise them for dairy products? How does the split of the two types of businesses in the authors' data set compared to that for the entire U.S. industry? Do the two types of business managers differ significantly in their decisions on breeding (e.g., for the most profitable animal genetics to sell vs. for most productive animals to raise) and use of genomic testing (e.g., to grab market share vs. to avoid inbreeding)?
2. Prior to the introduction of genomic testing, why were some lines of bulls more popular than others even though they had comparable or same genetic traits (page 15, top line)? Was the popularity based upon prices, actual productivity (as opposed to the measured genetic traits), branding effects, or something else? After the introduction of genomic testing, did any already popular lines (e.g., super-sires) become less popular and vice versa (because buyers found out some already popular lines now have higher inbreeding rates)? Is it possible for breeders to line breed different lines of cattle since 2008 as some lines ascended to super-sires while other lines became less popular? In other words, can

we guarantee there were no crossovers between the treatment group and the control group since 2008?

3. Section 2 Conceptual framework examined the two key, sequential decisions facing animal breeders: to line breed or cross breed; to have genomic testing or not. It would be more informative to establish a unified model that explicitly addresses the interdependence between these two decisions with the objective function of market share maximization (say, to breed animals with the highest trait values relative to the market). In this way, the authors can claim to explore the coordination between individual dairy breeders and the entire industry by analyzing the strategic actions of animal breeders.
4. Regarding the cost-benefit analysis in section 6, the authors admitted that this is only a rough, back-of-the-envelope calculation, although there are still potentially substantial benefits of genomic testing to be included. Once animal breeders decide to have their animals genomically tested, they can ascertain the animals' traits much sooner without waiting an additional three or four years for the daughter-proven results. Meanwhile, animal breeders need to pay about \$50-100 per animal for testing and possibly cull the undesirable animals at a loss. The adjusted net benefits of genomic testing for the entire industry are likely to be greater than originally estimated, which may render inbreeding a lesser issue in terms of the economic value.
5. I would also suggest adding more justifications when the authors try to project the estimated effects on inbreeding and net merit value using the authors' sample data into the U.S. dairy industry. Related to my comment #1 above, the adoption rates of genomic testing may vary between the (commercial) cattle breeders included in this study and cattle producers in the country, so the projected net benefits of genomic testing for the industry will have to be revised accordingly.
6. Some minor edits:
 - (1) Page 2, end of paragraph 1: "industr" should be "industry".
 - (2) Page 7, line 4 from bottom: a period is missing before Melton et al.
 - (3) Page 12, 2nd paragraph, line 3 from the paragraph end: add some numbers to justify "a substantial portion".
 - (4) Page 14, 1st paragraph of section 4, lines 4-5: Why are bulls that are NOT genomically tested more likely to have better quality traits and be more widely used?
 - (5) Page 16, line 4 from page bottom: Figure 5 in the main text is referred to here. Then drop "in the Appendix".
 - (6) Page 16, last sentence of the page: perhaps to conduct formal statistical tests on the similarities between the three distributions: matched, treat, and unmatched.
 - (7) Page 24, line 3 below Figure 8: "percent points" should be "percentage points".